Data-Driven Prediction of AI-Training Instability

Gradient excursions, range events, first-entry risk and telemetry-aware routing

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Abstract

This paper defines instability forecasts through explicit monitored boundaries rather than a generic notion of unhealthy training. It develops gradient-excursion and rolling-range targets, distinguishes endpoint risk from first-entry risk, and formulates causal WPM estimation, telemetry-aware routing and uncertainty. Proof trails establish normalization rules, hazard composition and the information limit of unpredictable future shocks. Exact quadratic examples explain why local dynamical instability is not identical to every telemetry excursion. The method preserves negative evidence: a structural representation is not a universal replacement for direct telemetry.

Research lineage and mathematical status. This study develops a WPM mathematical extension with reference to the author's published WPM architecture [1] and foundational n/s scaling-spectrum framework [2]. The n/s paper concerns conditional moment-spectrum parameterization; the WPM paper provides the broader architectural reference. The results proved here are mathematical consequences of stated assumptions, usually applications of standard linear algebra, probability or optimization. The WPM contribution is their organization into an explicit workflow. Modeling constructions and applicability conditions are identified separately; no universal forecasting superiority is asserted.

Predictor inputs. The forecasting layer described here is data driven: it learns from observed histories and matured outcomes. It is not fed the system's true governing equation or unobserved future states at issuance. Physical equations used below define observables, explain a limiting example or specify a separately labeled comparator. Optional supplied-map inverse problems and physics-informed variants must be declared separately; they are not evidence that the default forecast knows the generator.

1 Causal observation and resolution contract

Let $\mathcal{F}_t$ contain the observations actually reported by issuance t: loss measurements, gradient and update statistics, known schedules and verified context. A delayed validation measurement belongs to the information set only when reported. The forecast horizon P is fixed by the query; W selects past extent, and m is a declared temporal sampling or internal-resolution parameter. These variables need not be functions of one another. For each target T, issue a forecast $p_{s,P,r}$

using only $\mathcal{F}_s$, with lens $r = (W, m)$. Once the label has matured, its loss can enter a causal surface

$$\widehat{L}_t(r; T, P) = \frac{\sum_{s \in \mathcal{A}_t} w_s \ell_T(Y^T_{s,P}, p_{s,P,r})}{\sum_{s \in \mathcal{A}_t} w_s}. \tag{1.1}$$

The same matured set $\mathcal{A}_t$ is used in numerator and denominator. Extend maturity when persistence or reporting delay requires follow-up beyond $s + P$. Unresolved negative outcomes are not known negatives. Empty history is missing risk, not zero risk. A pocket is a connected low-risk region under a specified adjacency and threshold. Retain each pocket's flow cross-sections, position, existence, signed errors and identity history. Fixed uses an eligible reference lens; Multi aggregates same-target alternatives; Auto changes a specified policy with causal context and matured utility. A pocket minimum is not an automatic event probability. An interpretable candidate event model is

$$p_{t,P} = \operatorname{logit}^{-1}\left(\beta_0 + \beta^\top z_t + \sum_i \alpha_i \phi(\mathcal{P}_i) + \sum_{i<j} \gamma_{ij} \psi(\mathcal{P}_i, \mathcal{P}_j)\right). \tag{1.2}$$

The coefficients, feature maps and any calibration map must be fitted from permitted data. Pocket labels must not have an artificial ordinal meaning; use a permutation-invariant aggregation or preserve a well-defined relational graph. Shared observations create dependence among pockets and cannot be counted as extra independent measurements. The compact archived AI representation used observable location/scale proxies in its multi-system scenario comparison, not explicit fits of all n/s parameters. A full scaling-spectrum branch is an optional extension that must calculate and qualify its own descriptors. Structural location/scale features and an actually fitted n/s spectrum are not interchangeable descriptions.

2 An instability label must name its boundary

An optimizer may have mathematically unstable local dynamics, while a monitored gradient statistic exhibits only a transient excursion. A loss spike, exploding update norm, nonfinite value, constraint violation and validation deterioration are different events. Define one before fitting.

Let $G_t \geq 0$ be a reported gradient norm and M_t the median over a declared recent prefix window. One operational target is

$$R_{t,P} = \frac{\max_{1 \leq h \leq P} G_{t+h}}{M_t}, \qquad Y^{\text{inst}}_{t,P} = \mathbf{1}\{R_{t,P} > b_P\}, \quad M_t > 0. \tag{2.1}$$

The reference denominator is frozen at issuance. If $M_t = 0$ or is too uncertain, use a separately defined absolute boundary, a declared regularized denominator or abstention. A floor in the denominator changes the target and should not be hidden.

This maximum target asks whether an excursion occurs during the horizon, even if gradients return to normal at its endpoint. A label for G_{t+P} alone answers a different question. A persistence requirement such as d consecutive exceedances needs its own label and maturity time.

3 Units, relative scale and known context

Proposition 3.1 (Scale invariance of the ratio target)**.** *Multiplying every relevant gradient norm, including the issuance reference, by the same positive c leaves R unchanged. Adding a fixed denominator floor generally breaks that exact invariance unless the floor transforms too.*

Proof. The maximum and median both multiply by c, so c cancels in the quotient. With a floor ε, the denominator becomes $\max(cM_t, \varepsilon)$ rather than $c\max(M_t, \varepsilon)$ unless ε is scaled consistently. □

Relative normalization does not automatically make gradient norms comparable across architectures. Parameterization, loss reduction, batch size and gradient clipping can change their meaning. Include known settings or use a justified system map. A norm measured after clipping and one measured before clipping are different channels; the former can conceal an excursion of the latter.

Worked example 3.2 (A finite-horizon ratio)**.** With median recent norm 2 and future norms $(2, 3, 7, 2)$, R=7/2. At threshold 3, the event occurs even though the endpoint ratio equals 1. If gradient units are multiplied by 100, the ratio is unchanged when the denominator is transformed as well.

4 A rolling-range target is a separate construction

For a scalar observable Z and span w, define $A_t = \max_{t-w\le s\le t} Z_s - \min_{t-w\le s\le t} Z_s$. At issuance, freeze a boundary b_t and define $\tau = \inf\{h \ge 1 : A_{t+h} > b_t\}$. The probability of instability by P is $\mathbb{P}(\tau \le P \mid \mathcal{F}_t)$.

The future range window may include known pre-issuance values. That is part of the target definition, not leakage. Its unresolved future elements remain unavailable at issuance. A boundary re-estimated using future values would change the query and can create leakage.

Range instability can occur without a new cross-family coupling or any operator change. Therefore "zero coupling" is not automatically a negative instability label. Controls and interpretation must match the event rather than the name of a source model.

5 Local training dynamics as an analytical reference

For quadratic loss with symmetric positive definite Hessian H and fixed learning rate η, deterministic gradient descent is $\theta_{t+1} = (I - \eta H)\theta_t$. Orthogonal diagonalization gives multipliers $a_i = 1 - \eta\lambda_i$.

Theorem 5.1 (Quadratic stability boundary)**.** *Convergence for every initial state occurs exactly when $0 < \eta < 2/\lambda_{\max}$. An initially absent unstable eigenmode is not visible in that one exact trajectory until perturbed.*

Proof. Each eigenmode evolves as $a_i^h z_{i,t}$. All magnitudes decay for all initial conditions exactly when $|a_i| < 1$ for every i. A zero coefficient remains zero in noiseless linear dynamics regardless of its multiplier. □

This is a proof about a supplied local model. The data-driven forecast does not know H merely because the example displays it. A real optimizer may have momentum, stochastic gradients, nonquadratic curvature and changing data. Inferring a stability margin from telemetry is another statistical task with measurement and model uncertainty.

For a scalar unstable multiplier $|a| > 1$ and known current magnitude $|\theta_t| \le b$, the first strict crossing occurs at $h = \lfloor \log(b/|\theta_t|)/\log|a| \rfloor + 1$ when $\theta_t \ne 0$. At zero state the formula is undefined and the exact noiseless recurrence remains zero. Noise can seed growth and requires a stochastic model.

6 Endpoint and first-entry probabilities

Worked example 6.1 (Why endpoint probabilities miss instability)**.** Suppose the future path is $(1.2, 0.7)$ with probability $1/2$ and $(0.7, 0.7)$ with probability $1/2$, with boundary 1. Endpoint exceedance at step 2 has probability zero, while entry before or at step 2 has probability $1/2$.

Let $h_j = \mathbb{P}(\tau = j \mid \tau \geq j, \mathcal{F}_t)$ be a conditional hazard. Then

$$C_t(P) = 1 - \prod_{j=1}^{P} (1 - h_j). \tag{6.1}$$

Proof. Survival through each next time is conditional on survival so far. Multiplying these conditional survival probabilities is the chain rule. Subtract from one. □

No temporal independence is required. Replacing hazards by marginal gradient-exceedance probabilities is generally wrong. A direct classifier for the full maximum event is another valid construction, but independent classifiers for different P need not produce a coherent monotone horizon curve.

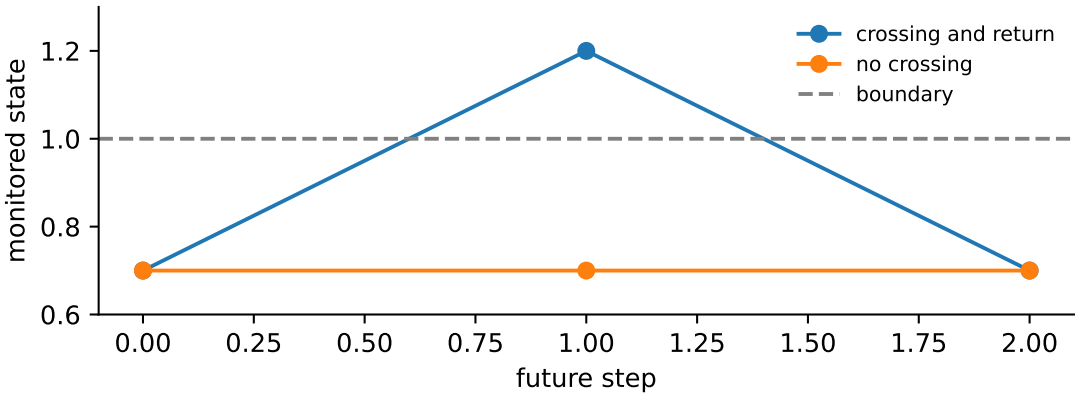


Figure 1: Two stipulated paths have the same endpoint and different boundary-arrival labels. An endpoint classifier cannot substitute for a first-entry model.

7 Telemetry and structural features have different roles

When the label is defined directly by future gradient norm, recent gradient history is a particularly relevant comparator. Structural loss location/scale features may summarize a different aspect of training and need not dominate. A combined predictor can exploit complementarity, but its parameters and regularization must be learned rather than assumed beneficial.

Theorem 7.1 (More information cannot worsen oracle Brier risk)**.** *Let* $p_X = \mathbb{E}[Y \mid X]$ *and* $p_{XZ} = \mathbb{E}[Y \mid X, Z]$ *for binary* Y*. The minimum expected Brier risk using* X *alone minus that using* (X, Z) *is* $\mathbb{E}[(p_{XZ} - p_X)^2] \geq 0$*.*

Proof. Decompose $Y - p_X = (Y - p_{XZ}) + (p_{XZ} - p_X)$. The cross term has zero expectation by conditioning on (X, Z). Taking squared expectations gives the result. □

This does not imply that adding estimated features improves a finite fitted model. Estimation error, model misspecification and overfitting can outweigh potential information. If Z is a deterministic pocket summary of X, the two sigma-fields contain the same information; practical benefit can still arise from representation and regularization, but not extra independent evidence.

8 An impossibility boundary for unforeseen shocks

If the target event is generated entirely by a future shock independent of $\mathcal{F}_t$ with probability p, then $\mathbb{P}(Y = 1 \mid \mathcal{F}_t) = p$. No telemetry or pocket function of that information can discriminate which individual shock will occur. The optimal probability is the base rate.

This follows directly from independence and the Brier decomposition. A future data corruption, hardware interruption or adversarial action outside the observation model can therefore defeat a seemingly detailed dynamical monitor. Measuring a relevant precursor can change the information set; relabeling existing pockets cannot.

9 Rare events, calibration and repeated warnings

For event prevalence p, a constant forecast p has Brier $p(1-p)$. At p=0.01, this equals 0.0099. A small Brier alone therefore does not establish useful event detection. If an evaluated subset has no positive events, its discrimination statistic is undefined or uninformative; its probability errors may still be assessed.

A fixed-horizon false-alarm probability does not directly control repeated alerting. Over K opportunities, the union bound gives probability of any false alert at most the sum of valid marginal bounds, regardless of dependence. Adaptive policies need those bounds to hold under their actual conditioning. Nominal calibration at each horizon is not automatically a sequential guarantee.

An action threshold follows a decision model. With one binary decision, false-alert cost c_A and missed-event cost c_M, warn when $p > c_A/(c_A + c_M)$. Repeated alerts, intervention costs and effects on the trajectory require a fuller policy model. Predictive instability risk alone does not identify the benefit of reducing learning rate.

10 Application worksheet and evidence scope

Choose a gradient excursion or range event, specify the reference window and denominator rule, and freeze its meaning. Fit telemetry, structural and combined same-target candidates with equal information cutoffs. Build matured risk landscapes separately for this label. Keep a qualified direct-telemetry reference available when the structural branch is weak or the relevant state is unrecognized.

For a current P=20 query, unknown future norms are the target, not features. Report probability, time units, context assumptions and whether the estimate concerns an endpoint, a maximum or sustained exceedance. A stopped or intervened run may be censored relative to the original no-intervention event and must not be silently treated as a negative.

Scoped evidence statement. The archived compact AI comparison did not establish structural superiority for gradient-defined instability; the specified direct-telemetry branch had stronger

discrimination [5]. A broad claim of closeness or replacement would be misleading for this target. Other AI targets and selected latent-failure comparisons were closer or favored structural features. The appropriate WPM rule is target-aware routing and a retained reference, not forced structural dominance.

11 Rules and solved questions

1. Declare the physical or operational instability event, including denominator and persistence rules.
2. Keep target thresholds independent of unresolved future outcomes.
3. Treat direct telemetry as a serious candidate when it measures the target's precursor.
4. Use hazards or joint paths for arrival risk; do not multiply marginal endpoint probabilities.
5. State censoring and policy assumptions, and allow fallback or abstention.

Exercise 11.1. If current norm reference is zero, is a future/reference ratio well-defined?

Solution. No. A declared alternative target or denominator policy is necessary; an undocumented numerical floor changes the event.

Exercise 11.2. Why can low event prevalence produce low Brier without useful discrimination?

Solution. The constant base-rate predictor already has loss $p(1-p)$, which is small when p is small. Ranking and useful warning behavior need separate examination.

Scope of the mathematical exposition

Unless explicitly identified as a sourced evidence statement, numerical values are constructed arithmetic examples. Application requires specifying the observation process, fitting unknown quantities, and checking reliability for the intended population and horizon. This study shares the AI prediction collection DOI 10.5281/zenodo.23784369.

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